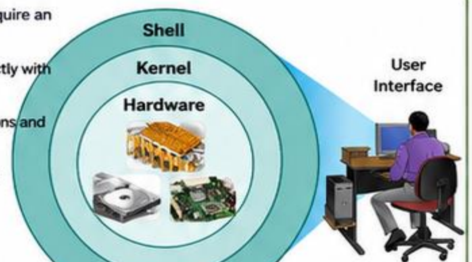
CISCO IOS ACCESS – KEY CONCEPTS

2.1.1 OPERATING SYSTEMS

- All end devices and network devices require an operating system (OS).
- The portion of the OS that interacts directly with computer hardware is called the kernel.
- The portion that interfaces with applications and the user is called the shell.
- The user interacts with the shell using a Command-Line Interface (CLI) or a Graphical User Interface (GUI).



Shell – The user interface that allows users to request specific tasks from the computer. Requests can be made through CLI or GUI.

Kernel – Communicates between the hardware and software of a computer and manages how hardware resources are used to meet software requirements.

Hardware – The physical part of a computer including underlying electronics.

GUI vs CLI

GUI (Graphical User Interface)

- Uses graphical icons, menus, and windows
- More user-friendly
- Requires less knowledge of command structure
- Example: Windows, macOS, Linux KDE, Apple iOS, Android
- Most users rely on GUI environments

CLI (Command-Line Interface)

- Text-based environment
- Requires knowledge of command structure
- Very little overhead
- Example:

```
analyst@secOps ~]$ ls
Desktop  Downloads
lab.support.files  second_drive
analyst@secOps ~]$
```

GUIs may not always provide all features, may fail or crash, or not operate as specified. Therefore, network devices are typically accessed through a CLI.

2.1.3 PURPOSE OF AN OS

Through a GUI, a PC operating system enables a user to:

- Use a mouse to make selections and run programs
- Enter text and text-based commands
- View output on a monitor

Through a CLI-based network operating system (e.g., Cisco IOS on a switch or router) enables a network technician to:

- Use a keyboard to run CLI-based network programs
- Use a keyboard to enter text and text-based commands
- View output on a monitor

- Cisco networking devices run particular versions of the Cisco IOS.
- The IOS version depends on the device type and required features.
- Devices come with a default IOS and feature set, but the IOS can be upgraded to obtain additional capabilities.

Cisco IOS Examples

Cisco Catalyst 2960 Switch
Runs a specific IOS version.

The family of network operating systems used on many Cisco devices is called the Cisco Internetwork Operating System (IOS).

- Used on many Cisco routers and switches of any type or size.
- Other IOS versions: IOS XE, IOS XR, NX-OS.

NOTE: The operating system on home routers is usually called firmware. The most common method for configuring a home router is by using a web browser-based GUI.

2.1.4 ACCESS METHODS

- A switch will forward traffic by default and does not need to be explicitly configured to operate. (Example: Two hosts connected to the same new switch can communicate.)
- Regardless of the default behavior of a new switch, all switches should be configured and secured.

METHOD	DESCRIPTION
Console	**Physical management port** that provides out-of-band access to a Cisco device. • Out-of-band access refers to access via a dedicated management channel used for device maintenance only. • Advantage: Device is accessible even if no networking services are configured (e.g., initial configuration). • Requires a computer running terminal emulation software and a special console cable.
Secure Shell (SSH)	**In-band and recommended** method for remotely establishing a secure CLI connection, over a network. • Uses a virtual interface over the network. • Requires active networking services on the device, including an interface with an IP address. • Most versions of Cisco IOS include an SSH server and an SSH client.
Telnet	**Insecure, in-band** method for remotely establishing a CLI session, over a network. • Uses a virtual interface over the network but does not provide a secure, encrypted connection. • Sends user authentication, passwords, and commands in plain text — only for lab environments. • Best practice is to use SSH instead of Telnet. • Cisco IOS includes both a Telnet server and Telnet client.

NOTE: Some devices, such as routers, may also support a legacy auxiliary (AUX) port to establish a CLI session remotely using a modem over a telephone connection. Similar to console, AUX is out-of-band and does not require networking services to be configured or available.

CISCO IOS NAVIGATION – PART 1

2.2.1 PRIMARY COMMAND MODES

The Cisco IOS separates management access into two primary command modes for security.

1 USER EXEC MODE

- Limited capabilities; useful for basic operations.
- Allows only a limited number of basic monitoring commands.
- Does NOT allow commands that change the device configuration.
- Identified by CLI prompt ending with the > symbol.

2 PRIVILEGED EXEC MODE

- Used to execute configuration commands.
- Full access to all commands and features.
- Can run monitoring commands and execute configuration and management commands.
- Higher configuration modes (like global config mode) can be reached from this mode.
- Identified by CLI prompt ending with the # symbol.

DEFAULT CLI PROMPTS

Command Mode	Description	Default Device Prompt

User EXEC Mode	• Mode allows access to only a limited number of basic monitoring commands.
	• It is often referred to as "view-only" mode.
		Examples:
		`Switch>`
		`Router>`

Privileged EXEC Mode	• Mode allows access to all commands and features.
	• The user can use any monitoring commands and execute configuration and management commands.
		Examples:
		`Switch#`
		`Router#`

2.2.2 CONFIGURATION MODE AND SUBCONFIGURATION MODES

To configure the device, the user must enter global configuration mode (also called global config mode).

GLOBAL CONFIGURATION MODE

- From global config mode, CLI changes are made that affect the operation of the device as a whole.
- Identified by a prompt that ends with `(config)#` after the device name.

Example: `Switch(config)#`

- Global configuration mode is accessed before other specific configuration modes.
- From global config mode, the user can enter different subconfiguration modes.
- Each subconfiguration mode allows the configuration of a particular part or function of the IOS device.

COMMON SUBCONFIGURATION MODES

1 LINE CONFIGURATION MODE

- Used to configure console, SSH, Telnet, or AUX access.
- Default prompt:

`Switch(config-line)#`

2 INTERFACE CONFIGURATION MODE

- Used to configure a switch port or router network interface.
- Default prompt:

`Switch(config-if)#`

NOTE

- The CLI prompt indicates the current mode.
- By default, every prompt begins with the device name.
- The remainder of the prompt tells you which mode you are in.

User EXEC Mode (Switch>) → Privileged EXEC Mode (Switch#) → Global Config Mode (Switch(config)#) → Subconfig Modes (line / if)

QUICK RECAP

Primary Command Modes

- User EXEC (>): limited, view-only
- Privileged EXEC (#): full access
- Higher modes are reached from privileged mode

Configuration Modes

- Global config mode: `Switch(config)#`
- Line config mode: `Switch(config-line)#`
- Interface config mode: `Switch(config-if)#`

CISCO IOS NAVIGATION – PART 2

2.2.4 Navigate Between IOS Modes

Move between IOS modes using specific commands. Privileged EXEC mode **(enable mode)** is the gateway to global configuration mode.

IOS Mode Navigation Commands

Action	Command	What it does
Enter Privileged EXEC mode (from User EXEC)	`enable`	Switch from user mode to privileged EXEC mode.
Exit Privileged EXEC mode	`disable`	Return to user EXEC mode.
Enter global configuration mode	`configure terminal`	Go to global configuration mode.
Exit global configuration mode	`exit`	Return to privileged EXEC mode.

Subconfiguration Modes

- Many subconfiguration modes exist (e.g., line, interface).
- Enter a subconfiguration mode with: **command + <>**
- Exit a subconfiguration mode with: **end**

Example: Line Configuration Mode

```
Switch(config)# line console 0
Switch(config-l ine)# exit
Switch(config)#
```

- Enters line configuration mode
- Exits back to previous mode

Key Combo

Ctrl+Z → Exit from subconfiguration mode (back to privileged EXEC mode).

Other Useful Commands

- **end** → Exit to pred EXEC mode.
- **interface FastEthernet 0/1** → Enter interface mode.

2.2.6 A Note About Syntax Checker Activities

What is Syntax Checker?

- A tool in Syntax Checker helps you check the correctenss of your commands.
- Stops you from entering wrong or incomplete commands.
- Uses real Syntax Checker in each mode.

How It Works

Enter a command in a specific mode → Syntax Checker validates it → Accepts the correct command

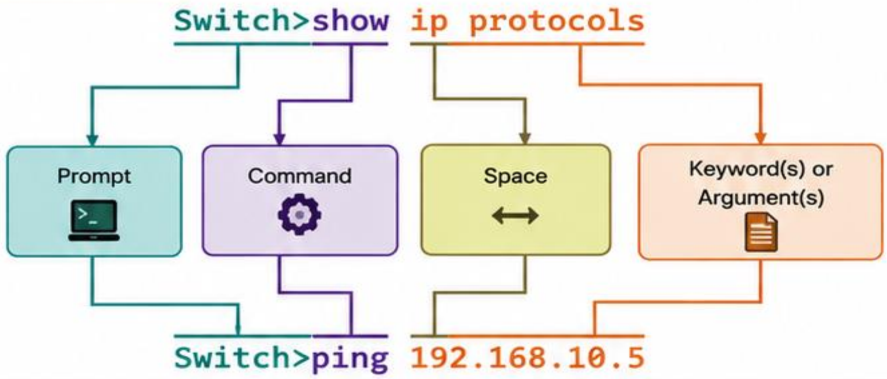
Important Points:

- Syntax **Checker** helps you enter **valid** and complete **Cisco IOS commands.**
- In each syntax checker, you must give the exact command structure.
- More advanced simulation tools (like Packet Tracer) use these commands in real equipment behavior.

CISCO IOS COMMAND STRUCTURE (PART 1)

2.3.1 BASIC IOS COMMAND STRUCTURE

- A Cisco IOS device supports many commands.
- Each command has a specific format (**syntax**) and can only be executed in the appropriate mode.
- General syntax: command followed by any appropriate **keywords** and arguments.



- **Keyword** – A specific parameter defined in the operating system. (Example: **ip protocols**)
- **Argument** – A value or variable defined by the user. (Example: 192.168.10.5)

ENTER

After entering each complete command, including any keywords and arguments, press **Enter** to submit the command to the command interpreter.

2.3.2 IOS COMMAND SYNTAX CHECK

- A command might require one or more arguments.
- The syntax provides the pattern or format that must be used when entering a command.
- The table shows the conventions used in Cisco **command syntax**.

Convention	Description
boldface	Boldface text indicates commands and keywords that you enter literally as shown.
italics	Italic text indicates **arguments** for which you supply values.
[x]	Square brackets indicate an optional element (keyword or argument).
{x}	Braces indicate a required element (keyword or argument).
{x | y | z}	Braces and vertical lines within square brackets indicate a required choice within an optional element. Spaces are used to clearly delineate parts of the command.

EXAMPLES

- **ping** *ip-address*
  Example: **ping 10.10.10.5**
- **traceroute** *ip-address*
  Example: **traceroute 192.168.254.254**

If a command is complex with multiple arguments, it may be represented like this:

```
Switch(config-if)# switchport port-security aging {  static | time time |    type {absolute |  inactivity)}
```

- The command will typically be followed with a detailed description of the command and each argument.
  The **Cisco IOS Command Reference** is the ultimate source of information for a particular IOS command.

2.3.3 IOS HELP FEATURES

Context-Sensitive Help

Enables you to quickly find answers to:

- Which commands are available in each command mode?
- Which commands start with specific characters or group of characters?
- Which arguments and keywords are available to particular commands?

To access: simply enter a question mark, **?**, at the CLI.

Command Syntax Check

- Verifies that a valid command was entered.
- When a command is entered, the CLI interpreter reads it from left to right.
- If the command is understood, the requested action is executed and the CLI returns to the appropriate prompt.
- If the command is not understood, the CLI provides feedback describing what is wrong with the command.

QUICK SUMMARY

- Use correct syntax: command + keywords + arguments
- Use help features: ? for help CLI checks syntax
- Press **Enter** to submit the command.

CISCO IOS COMMAND STRUCTURE – PART 2

2.3.5 HOT KEYS AND SHORTCUTS

- The IOS CLI provides hot keys and shortcuts that make configuring, monitoring and troubleshooting easier.
- Commands and keywords can be shortened to the minimum unique characters.
  - Example: **configure → conf** because **cu** is unique with **cu**
  - An even shorter version, **con**, will NOT work because more than one command begins with **con**.
  - Keywords can also be shortened.

A COMMAND LINE EDITING KEYS

Keystroke	Function
Tab — Tab	Completes a partial command name entry.
← Backspace	Erases the character to the left of the cursor.
Ctrl + D	Erases the character at the cursor.
Ctrl + K	Erases all characters from the cursor to the e o of the command line.
Esc + D	Erases all characters from the cursor to the end of the word.
Ctrl + U or Ctrl + X	Erases all characters from the cursor back to the beginning of the command line.
Ctrl + W	Erases the word to the left of the cursor.
Ctrl + A	Moves the cursor to the beginning of the line.
Left Arrow or Ctrl + B ←	Moves the cursor one character to the left.
Esc B	Moves the cursor back one word to the left.
Esc F	Moves the cursor forward one word to the right.
Right Arrow or Ctrl + F →	Moves the cursor one character to the right.
Ctrl E	Moves the cursor to the end of command line.
Up Arrow or Ctrl + P ↑	Recalls the previous command in the history buffer; beginning with the most recent command.
Down Arrow or Ctrl + N ↓	Goes to the next line in the history buffer.
Ctrl + R or Ctrl + I or Ctrl + L	Redisplays the system prompt and command line after a console message is received.

NOTE: While the **Delete** key typically deletes the character to the right of the prompt, the IOS command structure does not recognize the Delete key.

B --MORE-- PROMPT KEYS

When command output produces more text than can be displayed in a terminal window, the IOS will display a **"--More--"** prompt.

Keystroke	Function
Enter ↵	Displays the next line.
Space Bar	Displays the next screen.
Any other key *	Ends the display string and returns to previous prompt. * Except "y", which answers "yes" to the --More-- prompt, and acts like the **Space** bar.

C EXITING / ABORTING KEYS

Keystroke	Function
Ctrl + C	In any configuration mode, ends the configuration mode and returns to privileged EXEC mode. In setup mode, aborts back to the command prompt.
Ctrl + Z	In any configuration mode, ends the configuration mode and returns to privileged EXEC mode.
Ctrl + Shift + 6	All-purpose break sequence used to abort DNS lookups, traceroutes, pings, etc.

QUICK RECAP

Editing Keys
- ✔ Use shortcuts to move cursor quickly and edit commands.
- ✔ Examples: **Ctrl+A, Ctrl+E, Ctrl+W, Ctrl+K**

--More-- Prompt
- ✔ **Enter** = next line
- ✔ **Space** = next screen
- ✔ **Any other key*** = exit (* except "y")

Exiting Keys
- ✔ **Ctrl+C** = back to privileged EXEC
- ✔ **Ctrl+Z** = back to privileged EXEC
- ✔ **Ctrl+Shift+6** = break / abort

MEMORY TIP

Learn these shortcut keys – they save time and reduce errors while working in Cisco IOS CLI!

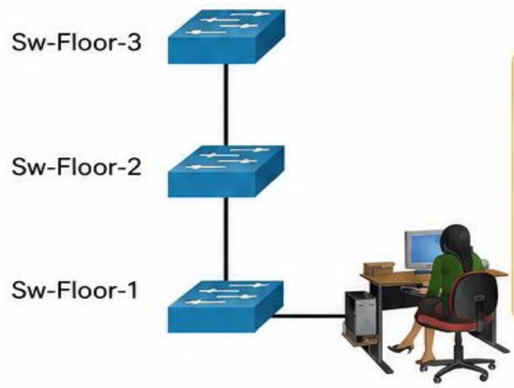
BASIC DEVICE CONFIGURATION (PART 1)

2.4.1 DEVICE NAMES

Give every device **a unique device name (hostname).**

GOOD HOSTNAME GUIDELINES

- Start with a letter
- No spaces
- End with a letter or digit
- Use only letters, digits, and dashes (-)
- Less than 64 characters



WHY NAME DEVICES?

- Easier to identify and manage devices
- Helpful for documentation
- Shows location & purpose
- Makes troubleshooting faster

APPLYING A DEVICE NAME (HOSTNAME)

- From privileged EXEC mode, enter global configuration mode:

```
Switch# configure terminal
Switch(config)# hostname Sw-Floor-1
Sw-Floor-1(config)#
```

Privileged EXEC `Switch#` → Global Configuration `Switch(config)#`

To return to the default prompt, use the **no hostname** global config command.

Always keep documentation updated each time a device is added or modified.

2.4.2 PASSWORD GUIDELINES

- Passwords protect devices and limit unauthorized access.
- Use strong passwords that are not easily guessed.
- Cisco IOS supports hierarchical passwords to allow different access privileges.
- Secure privileged EXEC, user EXEC, and remote Telnet access with passwords.

STRONG PASSWORD TIPS

- More than 8 characters
- Use a mix of **UPPERCASE**, lowercase, numbers, special characters, and/or numeric sequences
- Avoid using the same password for all devices
- Do not use common words (e.g., cisco, class)
- Use an Internet search to find a password generator
- Avoid weak passwords like: **cisco, class**

2.4.3 CONFIGURE PASSWORDS

You are in User EXEC mode by default. Console access is secured first. Then secure privileged EXEC and VTY access.

1 SECURE CONSOLE ACCESS

- Enter console line configuration mode.
- Set a password for console access.
- Enable user EXEC access with login.

```
Sw-Floor-1# configure terminal
Sw-Floor-1(config)# line console 0
Sw-Floor-1(config-line)#
 password cisco
Sw-Floor-1(config-line)# login
Sw-Floor-1(config-line)# end
Sw-Floor-1#
```

Console access will now require a password before entering user EXEC mode.

2 SECURE PRIVILEGED EXEC ACCESS

- Gain privileged EXEC mode access.
- Set an encrypted enable secret (stronger than enable password).

```
Sw-Floor-1# configure terminal
Sw-Floor-1(config)#
 enable secret class
Sw-Floor-1(config)# exit
Sw-Floor-1#
```

Enable secret is the most important security for privileged EXEC access.

3 SECURE VTY (REMOTE) ACCESS

- Enter VTY line configuration mode (0 to 15).
- Set a password for VTY access.
- Enable user EXEC access with login.

```
Sw-Floor-1# configure terminal
Sw-Floor-1(config)# line vty 0 15
Sw-Floor-1(config-line)#
 password cisco
Sw-Floor-1(config-line)# login
Sw-Floor-1(config-line)# end
Sw-Floor-1#
```

VTY lines allow remote access using Telnet or SSH. (Up to 16 lines: 0–15)

QUICK SUMMARY

Device Names
- Use meaningful hostnames
- Follow naming guidelines
- Use: `hostname <name>`

Passwords
- Use strong passwords
- Limit admin access
- Prefer enable secret

Configuration Steps
1. Secure console access
2. Secure privileged EXEC (enable secret)
3. Secure VTY access for remote login

BASIC DEVICE CONFIGURATION (PART 2)

2.4.4 ENCRYPT PASSWORDS

Why Encrypt Passwords?

- Startup-config and running-config files display most passwords in plain text.

➜ Anyone with access can view them.

What Does It Do?

- The **service password-encryption** command encrypts (obfuscates) all unencrypted passwords.
- Applies only to passwords in the configuration file, not to passwords as they travel over the network.
- Helps keep unauthorized users from viewing passwords in the file.

Key Points

- ✓ Weak encryption is used (type 7).
- ✓ Prevents casual viewing of passwords in the configuration file.
- ✓ Use **show running-config** to confirm passwords are encrypted.

ENCRYPT PASSWORDS – COMMAND

```
Sw-Floor-1# configure terminal
Sw-Floor-1(config)# service password-encryption
Sw-Floor-1(config)#
```

VERIFY – CHECK ENCRYPTION

```
Sw-Floor-1(config)# end
Sw-Floor-1# show running-config
!
(Output omitted)
!
line con 0
 password 7 094F471A1A0A
 login
!
line vty 0 4
 password 7 094F471A1A0A
 login
!
line vty 5 15
 password 7 094F471A1A0A
 login
!
end
```

2.4.5 BANNER MESSAGES

Why Use Banner Messages?

- Declares that only authorized personnel should access the device.
- Important for legal and security reasons.
- Some legal systems require visible notification before access.

Important Notes

- The **#** is the delimiting character.
- It can be any character or **symbol** as long as it does not appear in the message.
- After the command is executed, the banner is displayed on all subsequent attempts to access the device until removed.

BANNER MESSAGE – COMMAND SYNTAX

```
banner motd # message of the day #
```

Delimiting character (first #) — Delimiting character (second #)

EXAMPLE – CONFIGURE BANNER ON Sw-Floor-1

```
Sw-Floor-1# configure terminal
Sw-Floor-1(config)# banner motd #Authorized Access Only#
Sw-Floor-1(config)#
```

QUICK SUMMARY

Encrypt Passwords

- ✓ Use **service password-encryption** in global configuration mode.
- ✓ Encrypts passwords in the configuration file (not during transmission).
- ✓ Verify with **show running-config** (look for “password 7 ...”).

Banner Messages

- ✓ Use **banner motd #message#** command.
- ✓ Informs users before they access the device.
- ✓ Choose a delimiting character that does not appear in the message.
- ✓ Displayed until the banner is removed.

SAVE CONFIGURATION

2.5.1 CONFIGURATION FILES

Cisco devices use **two system files** to store configuration:

startup-config
Stored in NVRAM.
Contains the commands used when the device starts or reboots.
Flash does not lose its contents when the device is powered off.

running-config
Stored in RAM.
Reflects the current configuration.
Changes take effect immediately.
RAM is volatile – all content is lost when the device is powered off or restarted.

To view the current configuration stored in RAM, use the **show running-config** privileged EXEC command.

```
Sw-Floor-1# show running-config
Building configuration...
Current configuration : 1351 bytes
!
! Last configuration change at 00:01:20 UTC Mon Mar 1 1993
!
version 15.0
no service pad
service timestamps debug datetime msec
service timestamps log datetime msec
service password-encryption
!
hostname Sw-Floor-1
!
(output omitted)
```

To view the startup configuration stored in NVRAM, use **show startup-config** privileged EXEC command.

To save changes made to the running configuration to the startup configuration, use **copy running-config startup-config** privileged EXEC command.

2.5.2 ALTER THE RUNNING CONFIGURATION

RESTORE PREVIOUS CONFIGURATION

- If changes in the running-config do not have the desired effect and have not been saved, you can restore the device to its previous configuration.
- Remove changed commands individually, or reload the device using the **reload** command.

DOWNSIDE OF RELOAD

The device will be offline for a brief period, causing network downtime.

PROMPT ON RELOAD

- When a reload is initiated, the IOS detects unsaved changes in the running-config.
- A prompt appears asking whether to save changes.

TO DISCARD CHANGES

At the prompt, enter: **n** or **no**

ERASE STARTUP CONFIG

- If undesired changes were saved to the startup config, you may need to clear all configurations.
- Erase the startup-config and restart the device.

COMMAND

```
Sw-Floor-1# erase startup-config
```

CONFIRM ERASE

- After the command is issued, the switch will prompt you to confirm.
- Press **Enter** to accept and proceed.

PROMPT EXAMPLE

Erasing the nvram filesystem will remove all configuration files! Continue? [confirm]

AFTER ERASING AND RELOAD

- After removing the startup-config from NVRAM, reload the device.
- On reload, the switch loads the default startup configuration that originally shipped with the device.

RESULT

Device boots with default configuration.
No saved custom configurations remain.

QUICK SUMMARY

FILES
- **startup-config** → in NVRAM (saved across reboots)
- **running-config** → in RAM (active, lost on power off)

VIEW COMMANDS
- **show running-config** → see current (RAM) config
- **show startup-config** → see saved (NVRAM) config

SAVE CHANGES
- **copy running-config startup-config** → saves current config to NVRAM

PORTS AND ADDRESSES

2.6.1 IP ADDRESSES

WHY IP ADDRESSES?

- Primary way to locate devices
- Enables end-to-end communication
- Every device on a network must be configured with an IP address

EXAMPLES OF END DEVICES

- Computers, laptops, servers
- Network printers
- VoIP phones
- Security cameras
- Smart phones
- Mobile handheld devices (barcode scanners, etc.)

IPv4 (Internet Protocol Version 4)

- IPv4 address uses dotted decimal notation
- Four decimal numbers (0–255) separated by dots
- A subnet mask is required (32-bit)
- Example configuration:

Internet Protocol Version 4 (TCP/IPv4) Properties ×

General

○ Obtain an IP address automatically
◉ Use the following IP address:

IP address:	192 . 168 . 1 . 10
Subnet mask:	255 . 255 . 255 . 0
Default gateway:	192 . 168 . 1 . 1

○ Obtain DNS server address automatically
◉ Use the following DNS server addresses:

Preferred DNS server:	. . .
Alternate DNS server:	. . .

In the example:
IP: 192.168.1.10
Subnet Mask: 255.255.255.0
Default Gateway: 192.168.1.1

IPv6 (Internet Protocol Version 6)

- IPv6 address uses hexadecimal notation
- 128 bits written as a string of hex values
- Groups of four hex digits separated by colons (:)
- Not case-sensitive (a–f or A–F)
- Example configuration:

Internet Protocol Version 6 (TCP/IPv6) Properties ×

General

○ Obtain an IPv6 address automatically
◉ Use the following IPv6 address:

IPv6 address:	2001:db8:acad:10::10
Subnet prefix length:	64
Default gateway:	fe80::1

○ Obtain DNS server address automatically
◉ Use the following DNS server addresses:

Preferred DNS server:	
Alternate DNS server:	

In the example:
IPv6 Address: 2001:db8:acad:10::10
Prefix Length: 64
Default Gateway: fe80::1

KEY POINTS

- IPv4 addresses are 32-bit; IPv6 addresses are 128-bit.
- IPv4 uses dotted decimal; IPv6 uses hexadecimal.
- IPv4 is the most recent version and replaced IPV4.
- Every device must have a unique IP address.
- Subnet mask (IPv4) defines the network portion.
- Default gateway is the router used to reach other networks.

2.6.2 INTERFACES AND PORTS

WHAT ARE INTERFACES AND PORTS?

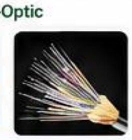
- Network communication depends on:
  - End device interfaces
  - Networking device interfaces
  - Cables that connect them
- Each interface has standards that define it.
- Cable must match the interface standards.

TYPES OF NETWORK MEDIA

Copper (Twisted-Pair)

Coaxial

Fiber-Optic



Wireless

MEDIA DIFFERENCES

Different media types have different characteristics:

- Distance a signal can travel
- Environment (indoor/outdoor, etc.)
- Amount of data & speed
- Cost and installation

IMPORTANT NOTES

- Not every media type works in every situation.
- Choose the right media for the purpose.
- More than one type can be used in a network.

NETWORKING DEVICE INTERFACES

- **Example:** Cisco Layer 2 switches (most common).
- Ports support Layer 3 IP addresses.
- Switch ports do NOT need IP addresses to forward packets.

SWITCH VIRTUAL INTERFACES (SVIs)

- Virtual interfaces used to manage switches over a network (IPv4/IPv6).
- One or more SVIs appear "out-of-the-box."
- Default SVI is interface VLAN 1 (default configuration).

NOTE

A Layer 2 switch does not need an IP address. The IP address assigned to the SVI is used to remotely access the switch. It is not required for the switch to perform its operations.

CONFIGURE IP ADDRESSING

2.7.1 MANUAL IP ADDRESS CONFIGURATION FOR END DEVICES

Purpose: Assign a static IP address so the end device can communicate with other devices on the network.

How to configure IPv4 manually on Windows:

1. Open **Control Panel → Network and Sharing Center → Change adapter settings.**
2. Right-click your adapter and select **Properties.**
3. Select **Internet Protocol Version 4 (TCP/IPv4)** and click **Properties.**
4. Select **Use the following IP address.**
5. Enter **IP address, Subnet mask,** and **Default gateway.**

Internet Protocol Version 4 (TCP/IPv4) Properties

- ○ Obtain an IP address automatically
- ◉ Use the following IP address:

IP address:	192 . 168 . 1 . 10
Subnet mask:	255 . 255 . 255 . 0
Default gateway:	192 . 168 . 1 . 1

Note: IPv6 addressing and configuration options are similar to IPv4.

2.7.2 AUTOMATIC IP ADDRESS CONFIGURATION FOR END DEVICES

DHCP

- Most end devices use DHCP (Dynamic Host Configuration Protocol) to get IP settings automatically.
- Saves time and reduces configuration errors.
- DHCP assigns: IP address, Subnet mask, Default gateway, and DNS server.

How to configure DHCP on Windows:

1. Select **Obtain an IP address automatically.**
2. Select **Obtain DNS server address automatically.**
3. Click **OK.**

Internet Protocol Version 4 (TCP/IPv4) Properties

- ◉ Obtain an IP address automatically
- ○ Use the following IP address:

IP address:	. . .
Subnet mask:	. . .
Default gateway:	. . .

- ◉ Obtain DNS server address automatically
- ○ Use the following DNS server addresses:

Preferred DNS server:	. . .
Alternate DNS server:	. . .

Note: IPv6 uses DHCPv6 and SLAAC (Stateless Address Autoconfiguration) for dynamic address allocation.

2.7.4 SWITCH VIRTUAL INTERFACE CONFIGURATION

To manage a switch remotely, configure an IP address on the Switch Virtual Interface (SVI). VLAN 1 is usually used.

Steps to configure an SVI (IPv4):

1. Enter global configuration mode.
2. Create/select VLAN (example: VLAN 1).
3. Assign IP address and subnet mask to the SVI.
4. (Optional) Set default gateway for remote networks.
5. Enable the SVI (no shutdown).

Example Configuration (Sw-Floor-1)

```
Sw-Floor-1# configure terminal
Sw-Floor-1(config)# interface vlan 1
Sw-Floor-1(config-if)# ip address 192.168.1.20 255.255.255.0
Sw-Floor-1(config-if)# no shutdown
Sw-Floor-1(config-if)# exit
Sw-Floor-1(config)# ip default-gateway 192.168.1.1
```

Note: Similar to Windows hosts, switches with an IPv4 address usually need a **default gateway** to reach remote networks. This is set using the **ip default-gateway** ***ip-address*** global command.

2.7.1 QUICK REFERENCE – IPV4 KEY POINTS

- IPv4 format: Dotted decimal (e.g., 192.168.1.10).
- Four decimal numbers (0–255) separated by dots.
- **Components:** IP address, Subnet mask, Default gateway (and DNS).
- **Manual config:** Useful for static or special devices.
- **Automatic config:** DHCP assigns all IP settings automatically.
- **Switch management:** Use SVI (VLAN interface) with IP address for remote access.

SUMMARY – CHOOSE THE RIGHT METHOD

END DEVICE NEEDS IP – To communicate on the network. → **CHOOSE METHOD** – • Manual (Static) IP • Automatic (DHCP) → **APPLY SETTINGS** – • Enter IP, Mask, Gateway • Or get via DHCP → **CONNECT & COMMUNICATE** – Device can now communicate with other devices.

KEY TAKEAWAYS

- IP addressing is essential for end-to-end communication.
- Use static IP for servers, printers or special needs.
- Use DHCP for most user PCs and mobile devices.
- Configure SVI and default gateway on switches for remote access.